

Gap Analysis & Audit Report: SIH 26003 MedTech Software

[!NOTE] This document audits the implemented CogniCare NER software against the original Smart India Hackathon (SIH) 26003 Problem Statement: "Development of software to be used as a MedTech Device for cognitive screening and interventions in older persons."

1. Problem Statement Requirements

Objective 1: Multi-domain Cognitive Assessment

- **Requirement:** Software must test working memory, episodic memory, executive functioning, sustained attention, language, and visuospatial functioning.
- **Implementation:**
 - **Memory:** Rhythm Recall (Bihu), Daily Routine Sort
 - **Executive & Visuospatial:** Bamboo Pattern Completion
 - **Language:** Word Association (Food)
 - **Attention:** Market Spot the Difference
 - **Memory (Episodic):** Gamosa Card Match
- **Gap: None.** All 6 required cognitive domains are actively tested through culturally adapted gamified interventions.

Objective 2: Real-time Adaptive Engine (CMAB)

- **Requirement:** Use Contextual Multi-Armed Bandits (CMAB) based on Thompson Sampling to adjust task difficulty dynamically.
- **Implementation:** The `CognitiveAdaptiveEngine (aiEngine.ts)` implements Thompson Sampling to balance exploration (trying new difficulties) and exploitation (serving the optimal difficulty). It calculates accuracy, speed, and standard deviation to adjust the reward distribution.
- **Gap: Minor.** The current implementation is a local, on-device bandit model. In a full production environment, the model's posterior distributions should be aggregated across all users centrally (federated learning) to improve the prior beliefs.

Objective 3: Flow Zone Maintenance

- **Requirement:** Keep the patient in the "flow zone" (optimal engagement without anxiety or boredom).
- **Implementation:** The AI engine specifically targets a 60-80% accuracy rate, dynamically scaling grid sizes, time limits, and memory item counts to maintain this sweet spot.
- **Gap: None.** The engine actively tracks and rewards sessions that fall within this accuracy band.

Objective 4: Culture-Specific Design

- **Requirement:** Ground the application in the cultural context of the targeted population (North East India).
- **Implementation:**
 - Visuals use NER palettes (Cream, Bark, Forest, Terracotta, Amber).

- Games use local contexts (Gamosas, Bihu rhythms, Assamese markets).
- Multi-language support (Assamese and English).
- **Gap: None.** The cultural integration is a core architectural pillar of the UI.

Objective 5: Multi-Role Portals

- **Requirement:** Interfaces for Patients, Caregivers, and ASHA workers.
- **Implementation:**
 - **Patient:** Gamified interface, Reminiscence Album, Sundowning calm protocols.
 - **Caregiver:** Analytics dashboard (Recharts), medication reminders, alerts.
 - **ASHA Worker:** Offline-first PWA screening portal for community deployment.
- **Gap: None.** Fully implemented.

2. Technical Implementation Evaluation

Feature	Status	Notes
Frontend Framework	✓	React + Vite + TailwindCSS. Fast and responsive.
Offline Capabilities	✓	vite-plugin-pwa installed and configured for ASHA workers in low-connectivity areas.
Analytics Visualization	✓	recharts implemented for Caregiver Radar and Line telemetry graphs.
State Management	✓	Zustand implemented for lightweight, scalable global state.
Database & Auth	⚠	GAP: Currently using mock data and local state. Requires Firebase or Supabase backend for persistent cloud storage and real-time syncing.
Native Mobile App	⚠	GAP: Currently a web-based PWA. For deeper device integrations (like native alarms, accelerometer data for tremors), a Flutter or React Native wrapper is recommended.

3. Recommended Next Steps (Phase 3)

[!IMPORTANT] To elevate this prototype to a production-ready medical device, the following steps are required:

1. **Cloud Backend Integration:** Connect the Zustand store to a secure, HIPAA/HIPAA-equivalent compliant database (e.g., Google Cloud Firestore) to persist patient telemetry.
2. **Federated AI Learning:** Move the Thompson Sampling priors to a cloud function, allowing the AI to learn from the aggregated population data rather than just individual patient history.
3. **Voice/Speech Analysis:** Integrate a Web Audio API / MediaRecorder module to analyze the *prosody* (rhythm and sound) of the patient's speech during the "Word Association" game, as speech changes are a strong early biomarker for dementia.

4. **Clinical Validation:** Deploy the prototype to a small test group at LGBRIMH Tezpur to gather empirical data on the CMAB engine's effectiveness.